

ON THE ELEMENTARY SYMMETRIC FUNCTIONS OF THE WISHART AND CORRELATION MATRICES

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1. Introduction

This paper is concerned with the elementary symmetric functions (esf's) of the Wishart matrix $nS(p \times p)$ with covariance Σ and $n(=N-1)$ degrees of freedom and correlation matrix $R = (s_{ij}/\sqrt{s_{ii}s_{jj}})$. The population correlation matrix is given by $P = (\sigma_{ij}/\sqrt{\sigma_{ii}\sigma_{jj}})$. In the next section confidence intervals for the esf's of Σ i.e. $\text{tr}_j \Sigma$, $j = 1, 2, \dots, p$ are derived. In section 3 the expected value of the zonal polynomial $C_K(RT)$, T any diagonal matrix, is derived and hence $E(\text{tr}_j R)$ only for the case $\Sigma = I_p$. In section 4 confidence intervals on $\text{tr}_j P$, $j = 1, 2, \dots, p$ are derived. In the last section simulation studies are done and some of the above confidence regions calculated.

2. On Confidence Regions of the esf's of Σ .

The expected value of $\text{tr}_j S$ is given by (de Waal (1972)).

$$\begin{aligned} 2.1 \quad E(\text{tr}_j S) &= (n(n-1) \dots (n-j+1)/(n^j)) \text{tr}_j \Sigma \\ &= (1 - \frac{1}{n})(1 - \frac{1}{n}) \dots (1 - \frac{j-1}{n}) \text{tr}_j \Sigma \end{aligned}$$

If n is large it is clear that $\text{tr}_j \Sigma$ can be estimated by $\text{tr}_j S$. To obtain confidence bounds on $\text{tr}_j \Sigma$ is not easy, but it is possible to derive approximate confidence intervals using the method of linearization. (Madansky and Olkin (1969)). The method assumes that

$$\text{tr}_j S \simeq \text{tr}_j \Sigma + \text{tr} \left(\frac{\partial \text{tr}_j \Sigma}{\partial \Sigma} (S - \Sigma) \right)$$

Hence (Crowther (1972))

$$2.2 \quad \text{Cov}(\text{tr}_j S, \text{tr}_k S) \simeq 2n^{-1} \text{tr}(H_j \Sigma H_k \Sigma) = h_{jk}(\Sigma)$$

where

$$2.3 \quad H_j = \frac{\partial \text{tr}_j \Sigma}{\partial \Sigma}$$

The approximate $100\gamma\%$ confidence interval for $\text{tr}_j \Sigma$ is given by (see Madansky and Olkin (1969))

$$2.4 \quad \text{tr}_j S \pm z_\gamma \sqrt{h_{jj}(S)}$$

where z_γ is the $\frac{100}{2}(1 + \gamma)\%$ point of the standard normal distribution.

The joint $100\gamma\%$ confidence region on $\text{tr}_j \Sigma$ $j = 1, 2, \dots, p$ will accordingly be given by

$$2.5 \quad (\hat{\tau} - \tau)' \Phi^{-1} (\hat{\tau} - \tau) \leq \chi_\gamma^2(p)$$

where

$$\tau' = (\text{tr}_1 \Sigma, \text{tr}_2 \Sigma, \dots, \text{tr}_p \Sigma), \hat{\tau}' = (\text{tr}_1 S, \text{tr}_2 S, \dots, \text{tr}_p S),$$

$$\Phi = (h_{jk}(S))_{j,k=1,2,\dots,p}$$

and χ_γ^2 the $100\gamma\%$ point of the Chi-square distribution with p degrees of freedom.

We need however H_j as defined in 2.3. The following lemma is useful to obtain this result:

Lemma 1 (Nel (1972)).

For any scalar λ and non-singular matrix $F(p \times p)$

$$\text{adj}(I_p - \lambda F) = B_0 + \lambda B_1 + \lambda^2 B_2 + \dots + \lambda^{p-1} B_{p-1}$$

where

$$B_r = \sum_{i=0}^r (-1)^i F^{r-i} \text{tr}_i F$$

Theorem 1. (A discussion with Nel made this theorem possible.)

For any non-singular $F(p \times p)$

$$\begin{aligned} \frac{\partial \operatorname{tr}_j F}{\partial F} &= (-1)^{j-1} \{ F^{j-1} - F^{j-2} \operatorname{tr}_1 F \\ &+ F^{j-3} \operatorname{tr}_2 F + \dots + (-1)^{j-1} I_p \operatorname{tr}_{j-1} F \} \end{aligned}$$

Proof

Since

$$\begin{aligned} \frac{\partial |F - \lambda I_p|}{\partial F} \\ = (-\lambda)^{p-1} \frac{\partial \operatorname{tr}_1 F}{\partial F} + (-\lambda)^{p-2} \frac{\partial \operatorname{tr}_2 F}{\partial F} + \dots + \frac{\partial \operatorname{tr}_p F}{\partial F} \end{aligned}$$

and

$$\begin{aligned} \frac{\partial |F - \lambda I_p|}{\partial F} \\ = (-\lambda)^{p-1} \operatorname{Adj}(I_p - \frac{1}{\lambda} F)' \\ = (-\lambda)^{p-1} B'_0 + (-\lambda)^{p-2} (-B'_1) + \dots + (-1)^{p-1} B'_{p-1} \end{aligned}$$

The result follows by comparing coefficients of $(-\lambda)^{p-j}$, i.e.

$$\frac{\partial \operatorname{tr}_j F}{\partial F} = (-1)^{j-1} B'_{j-1}$$

From the theorem the approximate estimate of the covariance between $\operatorname{tr}_j S$ and $\operatorname{tr}_k S$ will be given by

$$2.6 \quad h_{jk}(S) \simeq 2n^{-1} \{ \sum_{i=1}^j \sum_{r=1}^k (-1)^{(i+r)} \operatorname{tr}_{j-i} S \operatorname{tr}_{k-r} S \operatorname{tr} S^{i+r} \}.$$

If we consider

$$\operatorname{tr}_j Q_j S \simeq \operatorname{tr}_j Q_j \Sigma + \operatorname{tr} L_j (S - \Sigma)$$

where Q_j is an arbitrary $p \times p$ matrix and

$$2.7 \quad L_j = \frac{\partial \operatorname{tr}_j Q_j \Sigma}{\partial \Sigma},$$

then approximate confidence intervals for $\operatorname{tr}_j Q_j \Sigma$ can be obtained. A joint confidence region for $\operatorname{tr}_1 Q_1 \Sigma, \operatorname{tr}_2 Q_2 \Sigma, \dots, \operatorname{tr}_p Q_p \Sigma$ can also be obtained. We need:

Corollary 1

For any nonsingular $F(p \times p)$ and arbitrary matrix $Q(p \times p)$

$$\frac{\partial \operatorname{tr}_j QF}{\partial F} = (-1)^{j-1} Q' \Sigma \sum_{i=1}^j (-1)^{j-i} (F' Q')^{i-1} \operatorname{tr}_{j-i} QF$$

Proof

Since (Dwyer (1967))

$$2.8 \quad \frac{\partial h(Q_j F Q_k)}{\partial F} = Q_j' \frac{\partial h(Q_j F Q_k)}{\partial (Q_j F Q_k)} Q_k'$$

for a scalar function h of the matrix $Q_j F Q_k$, the corollary follows from theorem 1.

An approximate estimate of the covariance between $\operatorname{tr}_j Q_j S$ and $\operatorname{tr}_k Q_k S$ will therefore be given by

$$2.9 \quad 2n^{-1} \sum_{i=1}^j \sum_{r=1}^k \operatorname{tr}_{j-1} S Q_j \operatorname{tr}_{k-r} S Q_k \operatorname{tr}(S Q_j')^i (S Q_k')^r (-1)^{i+r}$$

Approximate confidence intervals can also be obtained for $\operatorname{tr}_j Q_j \Sigma^{-1}$. For this we need

$$\frac{\partial \operatorname{tr}_j Q_j \Sigma^{-1}}{\partial \Sigma}$$

Lemma 2

For any nonsingular $F(p \times p)$

$$2.10 \quad \frac{\partial \operatorname{tr}_j F^{-1}}{\partial F} = (-1)^{j-1} \sum_{i=1}^j (-1)^{j-i} (F^{-T})^{i+1} \operatorname{tr}_{j-1} F^{-1}$$

Proof

According to the notation by Dwyer (1967) we have that

$$\begin{aligned}
\frac{\partial \operatorname{tr}_j F^{-1}}{\partial F} &= \Sigma_{\gamma\delta} < \frac{\partial \operatorname{tr}_j F^{-1}}{\partial F^{-1}} > \frac{\partial < F^{-1} >}{\partial F} \\
&= \Sigma_{\gamma\delta} < (-1)^{j-1} \Sigma_{i=1}^j (-1)^{j-i} \operatorname{tr}_{j-i} F^{-1} (F^{-T})^{i-1} > . \\
&\quad F^{-T} K F^{-T} \\
&= (-1)^{j-1} \Sigma_{i=1}^j (-1)^{j-i} \operatorname{tr}_{j-i} F^{-1} (F^{-T})^{i+1}
\end{aligned}$$

where $K_{\gamma\delta}$ is a $p \times p$ matrix with 1 in the γ, δ -th position and 0 elsewhere and $< >_{\gamma\delta}$ indicates the γ, δ -th element of the matrix. F^{-T} is the transpose of F^{-1} .

Corollary 2

For any nonsingular $F(p \times p)$ and arbitrary matrix $Q(p \times p)$

$$2.11 \quad \frac{\partial \operatorname{tr}_j Q F^{-1}}{\partial F} = (-1)^{j-1} \Sigma_{i=1}^j (-1)^{j-i} \operatorname{tr}_{j-i} Q F^{-1} Q' (F^{-T} Q')^{i+1}$$

Proof

The result follows from lemma 2 by applying equation 2.7.

The estimate of the covariance between $\operatorname{tr}_j Q_j S^{-1}$ and $\operatorname{tr}_k Q_k S^{-1}$ will then be approximately given by

$$\begin{aligned}
2.12 \quad & 2n^{-1} \Sigma_{i=1}^j \Sigma_{r=1}^k (-1)^{i+r} \operatorname{tr}_{j-i} Q_j S^{-1} \operatorname{tr}_{k-r} Q_k S^{-1} . \\
& \operatorname{tr} Q_j (S^{-1} Q'_j)^{i+1} S Q'_k (S^{-1} Q'_k)^{r+1} S \\
& = 2n^{-1} \Sigma_{i=r}^j \Sigma_{r=1}^k (-1)^{i+r} \operatorname{tr}_{j-i} Q_j S^{-1} \operatorname{tr}_{k-r} Q_k S^{-1} \operatorname{tr} S^2 (S^{-1} Q'_j)^{i+2} \\
& \quad S^2 (S^{-1} Q'_k)^{r+2}
\end{aligned}$$

3. THE EXPECTED VALUES OF THE ESF'S OF THE CORRELATION MATRIX

The result in this section holds only for $\Sigma = I_p$, i.e. $P = I_p$.

The density of R is given by Anderson (1958) as

$$3.1 \quad ([\Gamma(\frac{1}{2}n)]^p / \Gamma_p(\frac{1}{2}n)) |R|^{1/2(n-p-1)}, \quad -1 < r_{ij} < 1.$$

The following theorem can be proved:

Theorem 2.

If R is distributed according to 3.1, then the expected value of the zonal polynomial $C_K(RT)$, T any diagonal matrix, corresponding to the partition

$K = (k_1, k_2, \dots, k_p)$ of k , is

$$3.2 \quad E C_K(RT) = (1/2n)_K C_K(T) / \prod_{i=1}^p (1/2n)_{k_i}$$

Proof

Constantine (1963) gave the following expressions for the zonal polynomials with $T = \text{diag}(t_1, t_2, \dots, t_p)$:

$$3.3 \quad C_K(ST) = d_{K,K} t_1^{k_1} t_2^{k_2} \dots t_p^{k_p} |S_1|^{k_1-k_2} |S_2|^{k_2-k_3} \dots |S_p|^{k_p} + \dots$$

$$3.4 \quad C_K(T) = d_{K,K} t_1^{k_1} t_2^{k_2} \dots t_p^{k_p} + \text{"lower terms"}$$

where

$$S_t = (s_{ij}), i, j = 1, 2, \dots, t$$

and $d_{K,K}$ a constant. By "lower terms" is meant all terms of a form similar to the one given, but for partitions $\tau < K$ of k . Further

$$3.5 \quad C_K(S) = d_{K,K} |S_1|^{k_1-k_2} |S_2|^{k_2-k_3} \dots |S_p|^{k_p} + \dots$$

Since (Constantine (1963)),

$$3.6 \quad \int_{S \succ 0} \text{etr}(-S) |S|^{1/2(n-p-1)} C_K(ST) dS \\ = \Gamma_p(1/2n, K) C_K(T)$$

we get after substitution of 3.3 and 3.4 respectively and comparing coefficients of

$$d_{K,K} t_1^{k_1} t_2^{k_2} \dots t_p^{k_p}$$

$$3.7 \quad \int_{D(R)} |R|^{1/2(n-p-1)} |R_1|^{k_1-k_2} |R_2|^{k_2-k_3} \dots |R_p|^{k_p} dR$$

$$= \Gamma_p(\frac{1}{2}n, K) / \Gamma(\frac{1}{2}n + k_1) \Gamma(\frac{1}{2}n + k_2) \dots \Gamma(\frac{1}{2}n + k_p).$$

$$= \{(\frac{1}{2}n)_K / \prod_{i=1}^p (\frac{1}{2}n)_{k_i}\} \{\Gamma_p(\frac{1}{2}n) / \Gamma^p(\frac{1}{2}n)\}$$

$$D(R) = \{-1 < r_{ij} < 1; i, j = 1, 2, \dots, p\} \text{ (see also Anderson (1958)).}$$

From 3.1 we have

$$\begin{aligned} 3.8 \quad E C_K(RT) &= ((\Gamma(\frac{1}{2}n))^p / \Gamma_p(\frac{1}{2}n)) \int_{D(R)} |R|^{\frac{1}{2}(n-p-1)} C_K(RT) dR \\ &= (E C_K(R) / C_K(I)) C_K(T) \end{aligned}$$

since the integral is symmetric in T . Using 3.3 and 3.4 respectively and comparing coefficients of

$$d_{K,K} t_1^{k_1} t_2^{k_2} \dots t_p^{k_p},$$

it follows from 3.7 that

$$\begin{aligned} (E C_K(R) / C_K(I)) &= ((\Gamma(\frac{1}{2}n))^p / \Gamma_p(\frac{1}{2}n)) (\Gamma_p(\frac{1}{2}n, K) / \\ &\quad \Gamma(\frac{1}{2}n + k_1) \Gamma(\frac{1}{2}n + k_2) \dots \Gamma(\frac{1}{2}n + k_p)) \\ &= (\Gamma(\frac{1}{2}n))^p (\frac{1}{2}n)_K / (\Gamma(\frac{1}{2}n + k_1) \Gamma(\frac{1}{2}n + k_2) \dots \Gamma(\frac{1}{2}n + k_p)) \\ &= (\frac{1}{2}n)_K / \prod_{i=1}^p (\frac{1}{2}n)_{k_i} \end{aligned} \quad \text{Q.E.D.}$$

This theorem is quite useful to obtain some moments of the esf's of the characteristic roots of R , such as the following:

We know that $\text{tr } R = p$ and from the theorem we have

$$E \text{tr } R = E C_1(R)$$

$$= p [\Gamma(\frac{1}{2}n)]^p (\frac{1}{2}n) / (\Gamma(\frac{1}{2}n + 1) (\Gamma(\frac{1}{2}n))^{p-1})$$

$$= p$$

$$\begin{aligned}
3.9 \quad E(\text{tr}_j R) &= \frac{j+1}{2^j} E C_{1j}^{(R)} \\
&= \frac{j+1}{j^j} [\Gamma(\frac{1}{2}n)]^p (\frac{1}{2}n)_{1j} C_{1j}^{(I_p)} ([\Gamma(\frac{1}{2}n+1)]^j [\Gamma(\frac{1}{2}n)]^{p-j}) \\
&= \binom{p}{j} [\Gamma(\frac{1}{2}n)]^j n(n-1)(n-2) \dots (n-j+1) / 2^j [\Gamma(\frac{1}{2}n+1)]^j \\
&= \binom{p}{j} n(n-1)(n-2) \dots (n-j+1) / n^j
\end{aligned}$$

From the density 3.1 we have

$$3.10 \quad E|R|^h = [\Gamma(\frac{1}{2}n)]^p \Gamma_p(\frac{1}{2}n+h) (\Gamma_p(\frac{1}{2}n) [\Gamma(\frac{1}{2}n+h)]^p,$$

If we substitute $j = p$ in 3.9, it follows that

$$\begin{aligned}
3.11 \quad E|R| &= [\Gamma(\frac{1}{2}n)]^p \Gamma_p(\frac{1}{2}n, 1^p) / (\Gamma_p(\frac{1}{2}n) [\Gamma(\frac{1}{2}n+1)]^p) \\
&= [\Gamma(\frac{1}{2}n)]^p \Gamma_p(\frac{1}{2}n+1) / (\Gamma_p(\frac{1}{2}n) [\Gamma(\frac{1}{2}n+1)]^p)
\end{aligned}$$

which serves also as a check for 3.9.

4. ON THE CONFIDENCE INTERVALS FOR THE ESF'S OF P

The density of R when $\Sigma \neq I_p$ seems to be very difficult to derive. Fisher (1962) gave the density in an integral form and could find explicit expressions only for special cases. To obtain the expected value of $\text{tr}_j R$ seems to be very difficult. It is, however, possible to derive approximate confidence intervals for $\text{tr}_j P$, $j = 2, 3, \dots, p$ if $P \neq I_p$. The method of linearization is again used by assuming

$$4.1 \quad \text{tr}_j R \simeq \text{tr}_j P + \text{tr } G_j(S - \Sigma)$$

where

$$G_j = \frac{\partial \text{tr}_j P}{\partial \Sigma}.$$

An approximation for the variance of $\text{tr}_j R$ will then be given by

$$\sigma^2(\text{tr}_j R) = E(\text{tr } G_j(S - \Sigma))^2, \quad j = 2, \dots, p$$

$$\begin{aligned}
&= E(\text{tr} G_j S)^2 - (\text{tr} G_j \Sigma)^2 \\
&= 2n^{-1} \text{tr}(G_j \Sigma)^2
\end{aligned}$$

The exact variance can however be derived from 3.2 if $P=I$, namely

$$\begin{aligned}
\sigma^2(\text{tr}_j R) &= E(\text{tr}_j R)^2 - (E \text{tr}_j R)^2 \\
&= ((j+1/2)^2 E(C_{1j}(R))^2 - (E \text{tr}_j R)^2
\end{aligned}$$

where

$$\begin{aligned}
E(C_{1j}(R))^2 &= \Sigma_D g_{1j,1j}^D E C_D(R) \\
&= \Sigma_D g_{1j,1j}^D \{ (\frac{1}{2}n)_D C_D(I_P) / \prod_{i=1}^P (\frac{1}{2}n)_{d_i} \}, D=(d_1, d_2, \dots, d_p).
\end{aligned}$$

D being a partition of $d=2j$ and $g_{1j,1j}^D$ a constant tabulated by Khatri and Pillai (1968) only for $d \leq 7$. It is therefore only possible to obtain the variances of $\text{tr}_2 R$ and $\text{tr}_3 R$ in this case.

To obtain G_j we need the following lemma:

Lemma 2

$$\text{Adj}(\Sigma - \lambda D(\Sigma)) = A_0 + \lambda A_1 + \lambda^2 A_2 + \dots + \lambda^{p-1} A_{p-1}$$

where

$$A_j = \prod_{i=1}^p \sigma_{ii}^{p-1} \Sigma^{-1} \Sigma_{k=0}^j (-1)^k P_D^{-(j-k)} \text{tr}_{p-k} P,$$

$$P_D^{-1} = D(\Sigma) \Sigma^{-1} \text{ and } D(\Sigma) = \text{diag}(\sigma_{11}, \sigma_{22}, \dots, \sigma_{pp}).$$

Proof

Assume

$$4.2 \quad \text{Adj}(\Sigma - \lambda D(\Sigma)) = A_0 + \lambda A_1 + \lambda^2 A_2 + \dots + \lambda^{p-1} A_{p-1},$$

then

$$4.3 \quad (\Sigma - \lambda D(\Sigma)) \text{adj}(\Sigma - \lambda D(\Sigma))$$

$$= |\Sigma - \lambda D(\Sigma)| I_p$$

$$= \Pi \sigma_{ii} |P - \lambda I| I_p$$

$$= \Pi \sigma_{ii} \{(-\lambda)^p + (-\lambda)^{p-1} \text{tr } P + (-\lambda)^{p-2} \text{tr}_2 P + \dots + \text{tr}_p P\} I_p.$$

But, substituting 4.2 in 4.3

$$4.4 \quad (\Sigma - \lambda D(\Sigma)) \text{Adj}(\Sigma - \lambda D(\Sigma))$$

$$= (\Sigma - \lambda D(\Sigma)) (\lambda^{p-1} A_{p-1} + \lambda^{p-2} A_{p-2} + \dots + A_0)$$

$$= (-\lambda)^p \{(-1)^{p-1} D(\Sigma) A_{p-1}\} + (-\lambda)^{p-1} \{(-1)^{p-1} \Sigma A_{p-1} + (-1)^{p-2} D(\Sigma) A_{p-2}\} + \dots + (-\lambda)^2 \{(-1)^2 \Sigma A_2 - D(\Sigma) A_1\}$$

$$- \lambda \{-\Sigma A_1 + D(\Sigma) A_0\} + \Sigma A_0.$$

Comparing coefficients of $(-\lambda)^{p-j}$ in 4.3 and 4.4. it follows that

$$A_j = (-1)^j \Pi \sigma_{ii} \Sigma^{-1} \text{tr}_{p-j} P + \Sigma^{-1} D(\Sigma) A_{j-1}$$

and therefore

$$4.5 \quad A_j = \Pi \sigma_{ii} \Sigma^{-1} \{(-1)^j \text{tr}_{p-j} P + (-1)^{j-1} P_D^{-1} \text{tr}_{p-j+1} P + \dots + P_D^{-j} \text{tr}_p P\}.$$

The following theorem can now be proved.

Theorem 3.

$$\frac{\partial \text{tr}_j P}{\partial \Sigma} = (-1)^j \Sigma^{j-1} (-1)^k \{P'_D {}^{-(j-k)} \Sigma^{-1} - D(P'_D {}^{-(j-k)} \Sigma^{-1})\} \text{tr}_{p-k} P$$

$$+ (\Sigma^{-1} - D^{-1}(\Sigma)) \text{tr}_{p-j} P.$$

Proof

As in theorem 1 $\frac{\partial \text{tr}_j P}{\partial \Sigma}$ will be the coefficient of $(-\lambda)^{p-j}$ in the expansion of

$$\frac{\partial |P - \lambda I|}{\partial \Sigma}$$

in a series of λ . Since

$$\begin{aligned} |P - \lambda I| &= \Pi \sigma_{ii}^{-1} |\Sigma - \lambda D(\Sigma)|, \text{ it follows that} \\ \frac{\partial |P - \lambda I|}{\partial \sigma_{kk}} &= \Pi \sigma_{ii}^{-1} \frac{\partial |\Sigma - \lambda D(\Sigma)|}{\partial \sigma_{kk}} + \Pi \sigma_{ii}^{-1} |\Sigma - \lambda D(\Sigma)| (-\sigma_{kk}^{-1}) \\ &= \Pi \sigma_{ii}^{-1} (1 - \lambda) (\text{Adj}(\Sigma - \lambda D(\Sigma)))'_{kk} \\ &\quad + \Pi \sigma_{ii}^{-1} |\Sigma - \lambda D(\Sigma)| (-\sigma_{kk}^{-1}) \end{aligned}$$

and

$$\frac{\partial |P - \lambda I|}{\partial \sigma_{\ell k}} = \Pi \sigma_{ii}^{-1} (\text{Adj}(\Sigma - \lambda D(\Sigma)))'_{\ell k}.$$

Hence

$$4.8 \quad \frac{\partial |P - \lambda I|}{\partial \Sigma} = \Pi \sigma_{ii}^{-1} [\text{Adj}(\Sigma - \lambda D(\Sigma))' - \lambda D(\text{Adj}(\Sigma))'] - |P - \lambda I| D^{-1}(\Sigma).$$

Using lemma 2, it follows that:

$$\begin{aligned} 4.9 \quad \frac{\partial |P - \lambda I|}{\partial \Sigma} &= \Pi \sigma_{ii}^{-1} A'_0 - D^{-1}(\Sigma) \text{tr}_P P \\ &\quad - \lambda [-\Pi \sigma_{ii}^{-1} (A'_1 - D(A'_0)) - D^{-1}(\Sigma) \text{tr}_{P-1} P] \\ &\quad + \lambda^2 [\Pi \sigma_{ii}^{-1} (A'_2 - D(A'_1)) - D^{-1}(\Sigma) \text{tr}_{P-2} P] + \dots \end{aligned}$$

Comparing coefficients of $(-\lambda)^j$ we have from 4.9 that

$$\begin{aligned} \frac{\partial \text{tr}_j P}{\partial \Sigma} &= (-1)^j \Pi \sigma_{ii}^{-1} (A'_j - D(A'_{j-1})) - D^{-1}(\Sigma) \text{tr}_{P-j} P \\ &= (-1)^j \sum_{k=0}^{j-1} (-1)^k (P'_D)^{-(j-k)} \Sigma^{-1} - D(P_D)^{-(j-k)} \Sigma^{-1} \\ &\quad \text{tr}_{P-k} P + (\Sigma^{-1} - D^{-1}(\Sigma)) \text{tr}_{P-j} P \quad \text{Q.E.D.} \end{aligned}$$

The approximate $100\gamma\%$ confidence interval for $\text{tr}_j P$ is therefore given by

$$4.10 \quad \text{tr}_j R \pm z_\gamma \delta(\text{tr}_j R)$$

where $\delta^2(\text{tr}_j R)$ is the approximate estimate of the variance of $\text{tr}_j R$, namely

$$4.11 \quad \partial^2(\text{tr}_j R) \simeq 2n^{-1} \text{tr}(S \frac{\partial \text{tr}_j R}{\partial S})^2$$

$$(S \frac{\partial \text{tr}_j R}{\partial S})^2 = \sum_{r=0}^j \sum_{k=0}^j w_r w_k$$

with

$$w_{k \neq j} = (-1)^k ((R_D^{-(j-k)} - D(R_D^{-(j-k)})) \text{tr}_{p-k} R$$

and

$$w_j = (I - R_D) \text{tr}_{p-j} R$$

where $R_D^{-1} = D(S)^{-1}$.

5. A SIMULATION STUDY FROM A NORMAL POPULATION

A normal population with $p = 5$ is considered. The population parameters are:

Population

$$\mu = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 1 & 1 & -1 & -1 & -1 \\ 1 & 2 & -1 & -1 & 0 \\ -1 & -1 & 2 & 2 & 2 \\ -1 & -1 & 2 & 3 & 3 \\ -1 & 0 & 2 & 3 & 5 \end{pmatrix}, \quad \begin{pmatrix} \text{tr}_1 \Sigma & = & 13 \\ \text{tr}_2 \Sigma & = & 40 \\ \text{tr}_3 \Sigma & = & 35 \\ \text{tr}_4 \Sigma & = & 11 \\ \text{tr}_5 \Sigma & = & 1 \end{pmatrix}$$

The characteristic roots are

$$(0,15 ; 2,88 ; 0,44 ; 0,58 ; 8,95)$$

A sample of $N = 50$ is drawn and the following statistics calculated:

Sample

$$\hat{\mu} = \begin{pmatrix} 0,09 \\ -0,14 \\ -0,04 \\ -0,27 \\ -0,67 \end{pmatrix}, \quad S = \begin{pmatrix} 0,96 & 1,08 & -0,83 & -0,83 & -0,85 \\ 1,08 & 2,01 & -0,80 & -0,93 & -0,09 \\ -0,83 & -0,80 & 1,76 & 2,02 & 2,11 \\ -0,83 & -0,83 & 2,02 & 3,05 & 3,11 \\ -0,85 & 0,09 & 2,11 & 3,11 & 5,23 \end{pmatrix}$$

$$\begin{pmatrix} \text{tr}_1 S & = & 13,0102 \\ \text{tr}_2 S & = & 39,5130 \\ \text{tr}_3 S & = & 34,1303 \\ \text{tr}_4 S & = & 9,6230 \\ \text{tr}_5 S & = & 0,7472 \end{pmatrix}$$

The characteristic roots of S are

$$(0,1263 ; 2,7816 ; 0,3193 ; 0,7363 ; 9,0467).$$

The covariance matrix of the esf's is calculated according to 2.6 and is given by

$$\Phi = \begin{pmatrix} 3,68 & 16,80 & 16,55 & 4,96 & 0,40 \\ 16,80 & 89,63 & 94,61 & 29,45 & 2,41 \\ 16,55 & 94,61 & 110,01 & 36,60 & 3,12 \\ 4,96 & 29,45 & 36,60 & 13,04 & 1,17 \\ 0,40 & 2,41 & 3,12 & 1,17 & 0,12 \end{pmatrix}$$

Using 2.4 the 95% confidence limits for $\text{tr}_j \Sigma$, $j = 1, 2, \dots, 5$ are calculated and are

$$\begin{array}{lll} 9,2486 & \leq & \text{tr}_1 \Sigma \leq 16,7718 \\ 20,9569 & \leq & \text{tr}_2 \Sigma \leq 58,0690 \\ 13,5726 & \leq & \text{tr}_3 \Sigma \leq 54,6881 \\ 2,5461 & \leq & \text{tr}_4 \Sigma \leq 16,6998 \\ 0,0825 & \leq & \text{tr}_5 \Sigma \leq 1,4120 \end{array}$$

The joint 95% confidence region is also considered. Using 2.5 it is found that

$$(\hat{\tau} - \tau)' \Phi^{-1} (\hat{\tau} - \tau) = 3,2644$$

which is less than $\chi^2_{0,95}(5) = 11,1$.

This simulation is repeated 36 times. Only in one case all the confidence limits do not cover the actual value; in 8 cases the confidence limits for $\text{tr}_5 \Sigma$ do not cover the actual value 1; in 3 cases the confidence limits for $\text{tr}_4 \Sigma$ do not cover the actual value 11; in 12 cases the calculated χ^2 -value is larger than the critical-value of 11,1.

An attempt was made to calculate the confidence limits for $\text{tr}_j P$ according to 4.1.1, but due to large rounding off errors, it was not possible to obtain any significant limits.

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